Week 2 Homework

Week 1 Homework

In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred () proof problems are eligible for inclusion in the proof portfolio assignment.

Exercises

Exercise

Sigma notation review: Write the sum in expanded form:

1. $\sum_{i=1}^{5} \sqrt{i}$
2. $\sum_{i=1}^{n} i^{10}$
3. $\sum_{j=0}^{n-1} (-1)^j$
4. $\sum_{j=n}^{n+3} j^2$

Exercise

Sigma notation review:Write the sum in sigma notation:

1. $1 + 2 + 3 + 4 + 5 + 6 + \cdots + 10$
2. $2 + 4 + 6 + 8 + \cdots + 2n$
3. $1 + x + x^2 + x^3 + \cdots + x^n$
4. $1 + 2x + 3x^2 + 4x^3 + \cdots + nx^{n-1}$

Exercise

Factorial review: What is the factorial function? What notation is used for this function?

Exercise

Factorial review: Calculate the following expressions using properties of the factorial function.

1. $\frac{7!}{3!}$
2. $\frac{8!}{4!4!}$
3. $\frac{10!}{9!}$
4. $\frac{n!}{(n-2)!}$

Exercise

Sequence Review: Write the first five terms of the following sequences:

1. $a_1 = 1$, $a_{n+1} = \frac{1}{1+a_n}$
2. $b_1 = 0$, $b_2 = 1$, $b_n = b_{n-1} - b_{n-2}$
3. $c_1 = 1$, $c_n = c_{n-1} + n$
4. $d_1 = 1$, $d_n = 2d_{n-1}$

Exercise

Sequence Review: Find a formula for the general term a_n of the given sequence:

1. $\{1, 4, 7, 10, 13 \cdots\}$
2. $\{-1, 2, -6, 24, -120 \cdots\}$
3. $\{1, 4, 9, 16, 25, \cdots\}$
4. $\{0, 1, 3, 7, 15, \cdots\}$